

Khoshrav Piran Doctor

Robotics Researcher

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Summary

Ph.D. candidate in Computer Science at UMass Amherst specializing in robotics and autonomous decision making systems operating under uncertainty. Experienced in building real-world robotic systems, developing learning-based models, and deploying scalable software solutions.

Education

University of Massachusetts Amherst , Computer Science Laboratory for Perceptual Robotics	Amherst, MA, USA 2020 – 2026
University of Massachusetts Amherst , Computer Science - Laboratory for Perceptual Robotics - Relevant Coursework: Robotics, Reinforcement Learning, Machine Learning, Artificial Intelligence, Feedback Control Systems	Amherst, MA, USA 2017 – 2020
K. J. Somaiya College of Engineering , Information Technology	Mumbai, India 2012 – 2016

Experience

Laboratory for Perceptual Robotics, UMass Amherst , Research Assistant - Designed intrinsically motivated methods for learning robot interaction dynamics (4× sample efficiency improvement) - Developed hierarchical skill learning approaches - Implemented control systems for a 3-DOF robotic head - Maintained and repaired the uBot robotic platform - Technologies: ROS, C++, Python, Deep Learning, Reinforcement Learning, SolidWorks	Amherst, MA, USA Aug 2017 – present 8 years 9 months
Platform for Ethical and Responsible Computing Education (PEaRCE) , Head Teaching Assistant - Led two teams to integrate ethics into CS curriculum (~1100 students) - Built tools to automate TA workflows and scale operations - Coordinated across students, instructors, and research staff - Led ethics discussion sessions across 5 courses - Improved website usability and infrastructure	Amherst, MA, USA Feb 2025 – Dec 2025 11 months
TRAC Labs, Inc. , Graduate Robotics Intern - Developed multi-agent systems for teleoperation situational awareness - Improved multi-agent path planning systems - Built RViz plugins and tools for better user interaction - Contributed to research design, writing, and documentation	Houston, TX, USA Sept 2021 – Aug 2023 2 years
Indiana University , Research Affiliate - Developed ML methods for detecting ASD and ADHD using IMU data - Designed neuroscience-inspired deep learning models (71% accuracy) - Published and presented research at neuroscience and physics venues	Bloomington, IN, USA Feb 2016 – present 10 years 2 months
AITOE Video Analytics Pvt. Ltd. , Software Developer - Built video analytics platform (search, summarization, statistics) - Developed security features (camera tampering, asset tracking) - Researched object tracking methods (CMT) - Collaborated with clients to define product requirements - Technologies: C++, NodeJS, Python, PHP, Caffe, QT	Mumbai, India July 2016 – July 2017 1 year 1 month

Computer Society of India, K. J. Somaiya College of Engineering Branch, Technical Head

- Developed content and trained council members to deliver workshops and seminars to the student body
- Responsible for management of the Technical Team of 4 individuals
- Responsible for the quality of the website and Android application of the Council
- Organized and delivered numerous workshops, seminars and technical competitions

Mumbai, India
July 2014 – June 2015
1 year 1 month

Team Eta, K. J. Somaiya College of Engineering, Electronics Head

Team Eta comprises of a group of students who create a fuel-efficient gasoline prototype vehicle annually to participate at the Shell Eco Marathon Asia. Team Eta ranked 5th in 2015 and was the highest ranked Indian team in its category.

- Responsible for the entire electronic system of the vehicle, including the Engine Control Unit, started motor and kill switch
- Designed and implemented an Arduino based GPS tracker for the vehicle to monitor key performance metrics

Mumbai, India
July 2014 – June 2015
1 year 1 month

Team Eta, K. J. Somaiya College of Engineering, Drive Train Team

- Researched possible technologies suited to deliver power and torque from the engine to the wheels
- Fabrication and assembly of the prototype

Mumbai, India
July 2013 – June 2014
1 year 1 month

Publications

Deep learning diagnosis plus kinematic severity assessments of neurodivergent disorders

Doctor, Khoshrav P, McKeever, Chaundy, Wu, Di, Phadnis, Aditya, Plawecki, Martin H, Nurnberger Jr, John I, Jose, Jorge V

www.nature.com/articles/s41598-025-04294-9

Deep Learning and Statistical longitudinal medication studies of attention-deficit/hyperactivity disorder of (ADHD) participants

Jose, Jorge V, Doctor, Khoshrav, McKeever, Chaundy Lee

schedule.aps.org/smt/2025/events/MAR-G65/1

A family of neurodevelopmental biomarkers extracted from a statistical analysis of kinematic data measured with high-definition sensors

Jose, Jorge, Doctor, Khoshrav, Phadnis, Aditya, McKeever, Chaundy, Plawecki, Martin

ui.adsabs.harvard.edu/abs/2023APS..MARA11013J/abstract

Deep Learning Diagnostic Analysis of the kinematic movement data recorded from individuals with Neurodevelopmental Disorders

Doctor, Khoshrav, Wu, Di, Phadnis, Aditya, Plawecki, Martin, Nurnberger Jr, John, Jose, Jorge

ui.adsabs.harvard.edu/abs/2022APS..MARD03003D/abstract

Learning Intrinsically Motivated Transition Models for Autonomous Systems

Doctor, Khoshrav, Ghosh, Hia, Grupen, Roderic

doi.org/10.1109/ICDL53763.2022.9962188

Evaluating Sensorimotor Abstraction on Curricula for Learning Mobile Manipulation Skills

Youngquist, Oscar, Spiro, Alenna, Doctor, Khoshrav, Grupen, Roderic

doi.org/10.1109/ICDL53763.2022.9962221

Statistical Motor Biomarkers Characterizing age-dependence in Neurodevelopmental Disorders

Doctor, Khoshrav, Wu, Di, Phadnis, Aditya, Nurnberger, John, Plawecki, Martin, Jose, Jorge

ui.adsabs.harvard.edu/abs/2021APS..MARS12015D/abstract

Demystifying multi-faceted video summarization: tradeoff between diversity, representation, coverage and importance

Kaushal, Vishal, Iyer, Rishabh, Doctor, Khoshrav, Sahoo, Anurag, Dubal, Pratik, Kothawade, Suraj, Mahadev, Rohan, Dargan, Kunal, Ramakrishnan, Ganesh
arxiv.org/abs/1901.01153

Learning From Less Data: A Unified Data Subset Selection and Active Learning Framework for Computer Vision

Kaushal, Vishal, Iyer, Rishabh, Kothawade, Suraj, Mahadev, Rohan, Doctor, Khoshrav, Ramakrishnan, Ganesh
doi.org/10.1109/WACV.2019.00142

Learning from less data: Diversified subset selection and active learning in image classification tasks

Kaushal, Vishal, Sahoo, Anurag, Doctor, Khoshrav, Raju, Narasimha, Shetty, Suyash, Singh, Pankaj, Iyer, Rishabh, Ramakrishnan, Ganesh
arxiv.org/abs/1805.11191

A unified multi-faceted video summarization system

Sahoo, Anurag, Kaushal, Vishal, Doctor, Khoshrav, Shetty, Suyash, Iyer, Rishabh, Ramakrishnan, Ganesh
arxiv.org/abs/1704.01466

PowerShare: A distributed processing system for Android devices

Kumar, Vignesh Pradeep, Jayawant, Shubham, Doctor, Khoshrav, Vidhani, Sanjay
doi.org/10.1109/CESYS.2016.7889989

Skills

Programming: C, C++, Python, Java, MATLAB, Arduino, Processing

Robotics & ML: ROS, Reinforcement Learning, Deep Learning, Planning, SolidWorks

Web Development: HTML, JavaScript, CSS, PHP

Databases: SQL, MongoDB

Interests

Robotics

Computational Neuroscience

Projects

Spatial Model Building Using a Robotic Head

July 2017 – Dec 2017

- Built probabilistic spatial models using deep learning
- Used models for goal-directed action selection

Chess Automated (Physical Chessboard)

July 2016

- Developed online and physical hybrid chess system
- Built Android control system for robotic board
- Debugged Arduino-based X-Y plotter mechanism

PowerShare: A Distributed Processing System for Android Devices

Jan 2015 – July 2016

- Developed distributed processing system for Android devices using Bluetooth and WiFi-Direct
- Implemented task scheduling and load balancing algorithms
- Evaluated system performance on real-world applications

Teaching

Teaching Assistant, University of Massachusetts Amherst	Aug 2017 – Dec 2025
• COMPSCI 220 (Fall 2017): Programming Methodology • COMPSCI 240 (Spring 2018): Reasoning Under Uncertainty • COMPSCI 503 (Fall 2018): Embedded Systems Engineering • COMPSCI 603 (Spring 2020): Robotics • COMPSCI 187 (Fall 2022): Programming with Data Structures • COMPSCI 683 (Spring 2023): Artificial Intelligence • COMPSCI 589 (Spring 2024): Machine Learning • Platform for Ethical and Responsible Computing (Fall 2024, Spring 2025, Fall 2025)	

Awards

Summer 2024 Dissertation Writing Fellowship	June 2024
Fall 2023 Proposal Writing Fellowship	Aug 2023
1st Place Project Competition	Mar 2016
2nd Place Project Competition	Mar 2016
Highest Committed Student Branch Activist	Dec 2014
Jewel of Somaiya Award for Outstanding Performance	Dec 2013
Cardinal Simon Pimenta Scholarship for Social Service	Dec 2008
Best Cadet Award of the N.C.C Naval Wing Troop	Dec 2009